Edge Computing

Limitations of Cloud-Only Architectures

- **High latency:** Centralized data centers introduce delays unsuitable for time-critical applications
- **Network dependency:** Full connectivity required; outages can halt operations
- **Bandwidth constraints:** Transmitting large volumes of data to the cloud is costly and inefficient
- **Privacy concerns:** Sensitive data may traverse multiple jurisdictions
- **Scalability limitations:** Centralized cloud may struggle with real-time, high-frequency workloads

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Definition of Edge Computing

Edge computing is a distributed computing paradigm in which computational tasks and data storage are performed closer to the sources of data, rather than solely in centralized cloud servers.

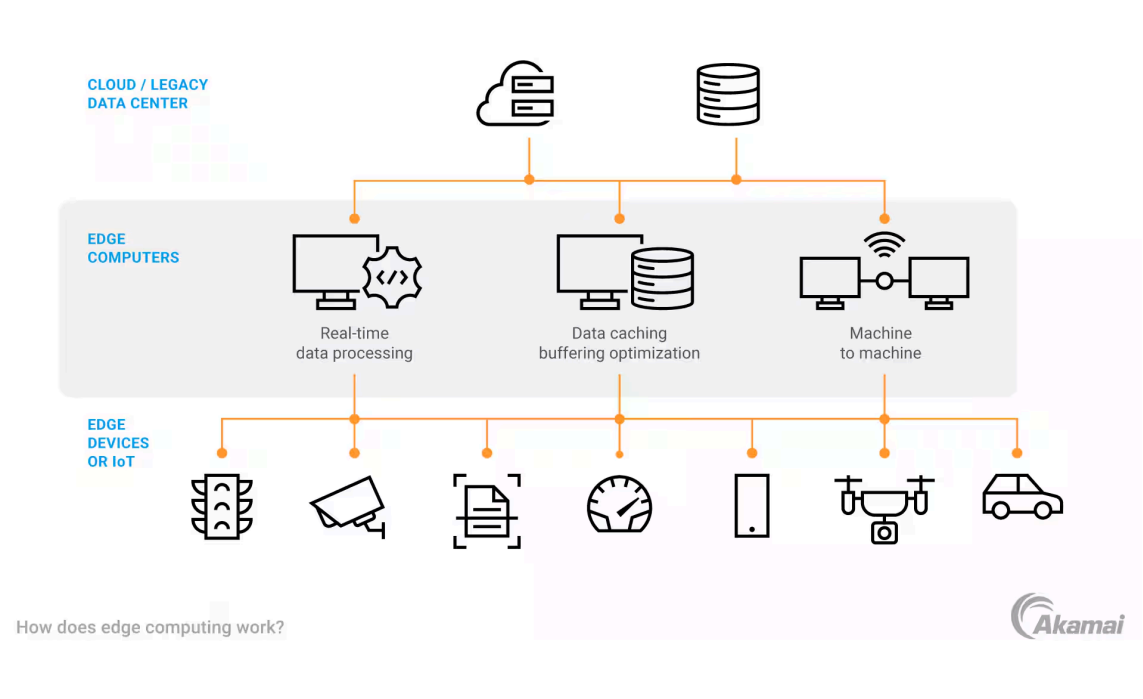
Deployment locations include:

- Industrial controllers and IoT devices
- Embedded systems and on-premise servers
- Micro data centers and regional/5G edge nodes

Edge-Cloud Complementarity

- **Edge:** Executes **time-critical**, **local**, **bandwidth-sensitive**, and **privacy-sensitive** operations
- **Cloud:** Performs **global coordination**, **long-term storage**, **machine learning training**, and **orchestration**

Key Benefit: This hybrid model reduces latency, conserves network bandwidth, enhances reliability, and preserves data privacy, while retaining the computational power and orchestration capabilities of the cloud.


CLOUD / LEGACY DATA CENTER
EDGE COMPUTERS
Real-time data processing
Data caching buffering optimization
Machine to machine
EDGE DEVICES OR IoT
Akamai

How does edge computing work?

Edge Computing is Inherently Distributed

Edge computing is a form of **distributed system**: instead of one centralized system, computation is spread across many geographically dispersed nodes. This naturally introduces classic distributed systems challenges:

Multiple Nodes Working Together

- remote communication
- state synchronization
- coordination mechanisms

Heterogeneity

- small embedded devices
- powerful on-prem servers
- cloud services

Intermittent Connectivity

- go offline
- operate in degraded network conditions
- sync periodically